

Introduction to Quantitative Political Analysis

GOVT 10: Quantitative Political Analysis

This chapter introduces quantitative political analysis: what it is, why political scientists reason with data, and how the scientific method structures empirical inquiry. It also gets you set up with the computing environment you will use all term.

1 Introduction to Quantitative Political Analysis

1.1 What Quantitative Political Analysis Is

Quantitative political analysis is the systematic use of numbers, data, and statistical methods to understand political phenomena. That sounds abstract, but the everyday applications are concrete. When a pollster reports that 47 percent of voters approve of the president, when a forecaster estimates that a Senate candidate has a 62 percent chance of winning, when a researcher claims that voter ID laws reduce turnout by two percentage points, all of them are doing quantitative political analysis. They are taking observations about the political world, summarizing them with numbers, and trying to reach conclusions that go beyond what any single person could see on their own.

This is how we test theories about how politics actually works, measure public opinion with some honesty about its uncertainty, evaluate whether policies do what they were designed to do, predict election outcomes, and make decisions based on evidence rather than gut feeling. None of these tasks require us to abandon political judgment. They require us to discipline that judgment with data.

1.2 Why It Matters

We live in an unprecedented age of political data. Every roll call vote in Congress is recorded. Every campaign donation above a small threshold is logged with the FEC. Polls run continuously. Social media generates a real-time stream of political talk. Precinct-level election returns are public. The question is no longer whether data shapes politics. It is whether you can read that data carefully enough to avoid being misled by it.

Even if you never work in politics professionally, these are skills of citizenship. You will be able to evaluate the statistical claims politicians make, understand what a poll actually tells you (and when its margin of error makes its headline meaningless), spot misleading visualizations in campaign ads, and read election forecasts without confusing a 70 percent chance with a certainty.

A Question to Hold Onto

Why should we use data and statistics to study politics at all? The short answer is that without them, we are stuck with intuition, and intuition about politics is notoriously unreliable. We remember dramatic stories and forget routine ones. We notice evidence that fits our prior beliefs. We confuse correlation with causation. Systematic data analysis is not a magic cure, but it is the best correction we have.

1.3 The Scientific Method in Political Research

Political science is, sometimes, a science. It earns that label when it follows the same basic method that other empirical sciences use:

1. **Observe** patterns in political behavior.
2. **Hypothesize** about what causes those patterns.
3. **Test** the hypothesis with data.
4. **Replicate** the test in different contexts and on different data.
5. **Revise** the theory when evidence calls for it.

This sequence is harder to apply to politics than to chemistry. We rarely get clean experiments. The actors in our data are people who respond to being studied. The questions we care about are politically charged. But the discipline of the method, of writing down a hypothesis before you look at the data and being willing to revise your beliefs when the evidence disagrees, is what makes political science science rather than punditry.

1.4 A Historical Example: John Snow and the Broad Street Pump

The clearest example of the scientific method beating intuition is not from political science at all. It is from public health, but the logic is exactly the logic of quantitative political analysis.

In late summer 1854, the Soho neighborhood of London was hit by a sudden and devastating cholera outbreak. More than 500 people died within ten days. The conventional wisdom of the time was that cholera spread through “bad air,” or miasma, rising from rotting organic matter. This was not a fringe view; it was the consensus of established medical authority.

Dr. John Snow, a London physician and early epidemiologist, disagreed. He suspected that cholera was transmitted through contaminated water, not through the air. But disagreement alone was not evidence. To overturn established medical doctrine, he needed something better than a competing theory. He needed systematic data.

What Snow did is the template for quantitative analysis everywhere. He went into Soho and mapped every death. He placed a mark on a street map for each cholera victim. Then he plotted every public water pump in the neighborhood. Then he looked for a pattern.

The pattern was striking. The deaths clustered tightly around the public pump on Broad Street. Houses near the Broad Street pump suffered hundreds of deaths. Houses near other pumps

suffered very few. A nearby brewery, where workers drank beer brewed with their own well water rather than the Broad Street water, recorded zero cholera deaths. A nearby workhouse with its own well had only five.

Snow brought this evidence to local officials. The handle of the Broad Street pump was removed. New cases of cholera in the neighborhood dropped almost immediately.

What John Snow Did, in Four Steps

1. **Systematic data collection.** Every death was recorded, not just the dramatic ones.
2. **Pattern recognition.** Mapping the deaths next to the pumps made the relationship visible.
3. **Evidence-based conclusion.** The data overturned a theory that "everyone knew" was right.
4. **Real-world action.** Removing the pump handle saved lives.

This is exactly the structure of careful political analysis: collect data systematically, look for patterns honestly, draw conclusions the data actually supports, and use those conclusions to inform action.

1.5 The Role of AI in Modern Political Analysis

The traditional way of teaching quantitative methods, where students memorize syntax and spend hours debugging punctuation, is increasingly outdated. AI tools, when used carefully, handle a lot of the mechanical work of writing code. That frees you to focus on the parts that actually require human judgment: deciding what question to ask, choosing an appropriate method, interpreting what the numbers mean politically, and noticing when something looks wrong.

This course uses AI deliberately. You will use a tool like Claude to help you write and debug R code. But you will also learn that AI is fallible. It can suggest that a correlation implies causation, miss political context that a human would recognize, recommend the wrong statistical method, or even hallucinate data that does not exist. Your job is to develop the judgment to evaluate what AI gives you.

AI Pitfalls to Watch For

- **Correlation vs. causation.** AI happily describes correlations as causal effects. They are not the same thing.
- **Missing context.** AI does not know that a "Republican-leaning district in 1980" means something very different from a "Republican-leaning district in 2024."
- **Biased data.** If a dataset systematically excludes certain voters, AI will not flag the problem for you.
- **Wrong methods.** AI may suggest a statistical technique that sounds reasonable but is inappropriate for your question.

- **Hallucination.** AI sometimes invents results, citations, or even datasets. Verify everything.

2 Using AI for This Material

Where AI Helps This Week

- Explaining an R error message line by line, including what a missing-package or missing-file error actually means.
- Translating base-R code you find online into the tidyverse equivalents we use in class.
- Describing in plain English what `library()`, `read_csv()`, `glimpse()`, and `mean()` are doing under the hood.

Where AI Gets This Wrong This Week

- It cannot see your folder structure, so it will confidently invent file paths or assume a working directory that does not match yours.
- It cannot diagnose Positron install or package-binary failures it cannot observe, and will guess at fixes that may make things worse.
- It tends to gloss over the `<-` versus `=` distinction as "stylistic" when this course treats `<-` as the assignment operator you should use.

A prompt that works for this week:

I ran `read_csv("turnout.csv")` and got Error: `'turnout.csv'` does not exist in current working directory. Explain what that error means in plain English and three things I should check before re-running.

A prompt that misleads:

Write the R code I need to load my data.

Why it misleads: AI does not know your dataset's filename, location, or columns, so it will produce code that runs cleanly on a fake file and fails on yours.

3 Common Mistakes and How to Avoid Them

Mistake 1: Forgetting to Load the Library

If `read_csv()` or `glimpse()` returns an error like `could not find function`,

the cause is almost always that you forgot to run `library(tidyverse)` at the top of your session. Make it a habit.

Mistake 2: Mismatched Types

Numbers stored as text will silently break math. If `mean()` returns an error, check `class()` on the column. A common cause is that one stray entry in a CSV (a header in the middle of the data, a "N/A" string) coerces the whole column to character.

Mistake 3: Confusing Correlation with Causation

The most important warning in this course, and one we will return to repeatedly. Two things that move together do not necessarily cause each other. Negative ads correlate with losing campaigns, but mostly because losing campaigns go negative when they are already behind. We will spend much of Chapter 4 on this.

4 R Functions Reference

Function	Purpose	Example
<code>library()</code>	Load a package	<code>library(tidyverse)</code>
<code>read_csv()</code>	Read a CSV into R	<code>read_csv("data.csv")</code>
<code>c()</code>	Create a vector	<code>c(1, 2, 3)</code>
<code>class()</code>	Check the data type	<code>class(x)</code>
<code>length()</code>	Count elements	<code>length(x)</code>
<code>mean()</code>	Calculate the average	<code>mean(x)</code>
<code>sum()</code>	Add values	<code>sum(x)</code>
<code>min()</code> , <code>max()</code>	Find extremes	<code>min(x)</code> , <code>max(x)</code>
<code>glimpse()</code>	Compact data overview	<code>glimpse(df)</code>
<code>head()</code>	First few rows	<code>head(df)</code>
<code>summary()</code>	Quick statistical summary	<code>summary(df)</code>
<code>table()</code>	Count by category	<code>table(x)</code>

Table 1: Core R Functions for Chapter 1

5 Practice Problems

Question 1

In your own words, explain what John Snow did that made his investigation an example of quantitative analysis rather than just educated guessing. What would have been different if Snow had simply argued from theory without collecting the map data?

Question 2

You write `mixed <- c(45, "Republican", TRUE)`. What type will `mixed` be, and why? What will happen if you then try `mean(mixed)`?

Question 3

Create a vector called `turnout` containing voter turnout rates for six countries: 67.5, 72.1, 58.3, 69.8, 71.2, 55.0. Compute its mean, median, minimum, and maximum. Which of those four numbers would you report as a description of "typical" turnout, and why?

Question 4

You load a dataset and run `glimpse()`. The column `vote_share` shows `<chr>` next to it, even though it looks like a percentage. What does this tell you about the column, and what is one thing that might have caused it?

For Further Study

The free online textbook *R for Data Science* by Hadley Wickham and Garrett Grolemund (at r4ds.hadley.nz) is the best introduction to R and the tidyverse. For the history of John Snow's investigation, Steven Johnson's *The Ghost Map* is a beautifully written narrative account. For a feel for how political scientists actually use quantitative methods, browse the abstracts in a recent issue of the *American Political Science Review*.